

User Acceptance Testing (UAT)

Date	29/09/2026
Team ID	SWTID-2026-7994
Project Name	AI StudyBuddy
Maximum Marks	-

Project Overview:

Project Name	AI StudyBuddy
Project Description	An AI-powered learning backend that lets students upload study material and generate summaries, flashcards, quizzes and personalized study plans using Google Gemini, secured with JWT and role-based access.
Project Version	1.0
Testing Period	To be filled by team

Testing Scope:

- Registration and login with JWT tokens
- Role-based access for Student and Admin
- Study material upload (.txt, .md, .pdf, up to 5 MB)
- AI summary, flashcards, quiz and study plan generation
- Saved resources and admin operations
- Error handling and validation

Testing Environment:

URL / Location	http://localhost:5000 (backend started with npm start), tested with Postman or Thunder Client
Credentials	Test accounts for the student and admin roles

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Register a new student	1. POST /api/auth/register with name, email, password, role	201 Created; success message with tokens and user details		
TC-002	Register with an existing email	1. Register the same email again	Request rejected with an error		
TC-003	Login with valid credentials	1. POST /api/auth/login with email and password	200 OK; access and refresh tokens returned		
TC-004	Login with wrong password	1. POST valid email with wrong password	Access denied with error message		
TC-005	Upload a valid study file	1. Send a .txt file as form-data to /api/materials/upload with token	201 Created; material saved with title and content		
TC-006	Upload an unsupported file	1. Upload a file of another type	Rejected: only .txt, .md and .pdf allowed		

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-007	Generate summary	1. POST /api/materials/:id/summarize	200 OK; concise summary returned		
TC-008	Generate flashcards	1. POST /api/materials/:id/flashcards	200 OK; list of question and answer flashcards		
TC-009	Generate quiz	1. POST /api/materials/:id/quiz	200 OK; questions with options and answers		
TC-010	Generate study plan	1. POST /api/materials/:id/study-plan with goal, hoursPerDay, days	200 OK; day-wise study plan returned		
TC-011	Access API without token	1. Call a protected route with no Authorization header	401 Not authenticated		
TC-012	Student calls an admin route	1. Log in as student 2. Call /api/admin route	403 Admins only		

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001					
BG-002					
BG-003					

Sign-off:

Tester Name	
Date of Test Completion	
Signature	

Notes:

- Cover both positive and negative scenarios in all test cases.
- Testers should give detailed feedback and suggestions for improvement.
- Bug tracking should include severity, status and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before deployment.